Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix

Source: Our World in Data | Data: Ember, Energy Institute (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse gas emissions originate from fossil fuels and industry

Fossil fuels have dominated energy systems since the Industrial Revolution, driving both climate change and significant health impacts.

CO2 Greenhouse gas emissions driving climate change

PM 5+ million premature deaths annually from air pollution

FF Fossil fuels dominate energy systems since the Industrial Revolution

Source: Our World in Data, Global Carbon Project, WHO

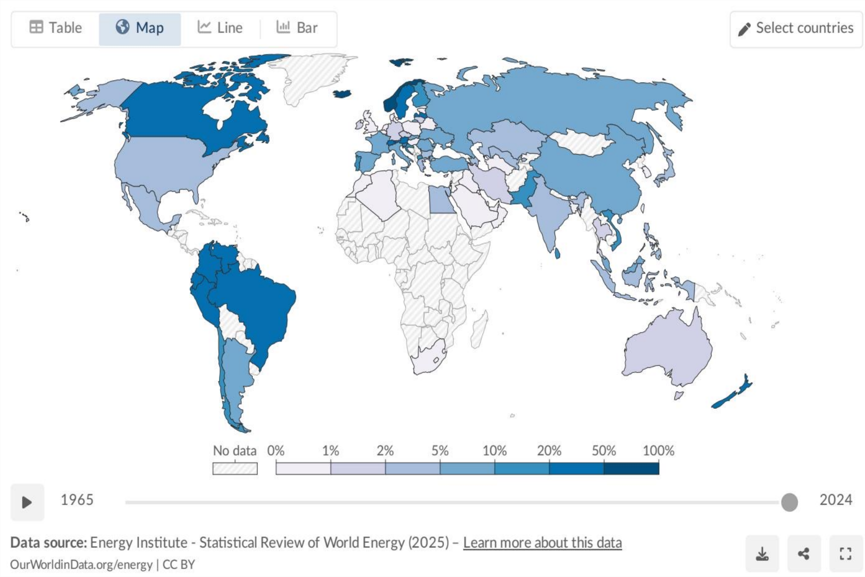
Renewables Now Supply ~14% of Global Primary Energy

14%

~1/7

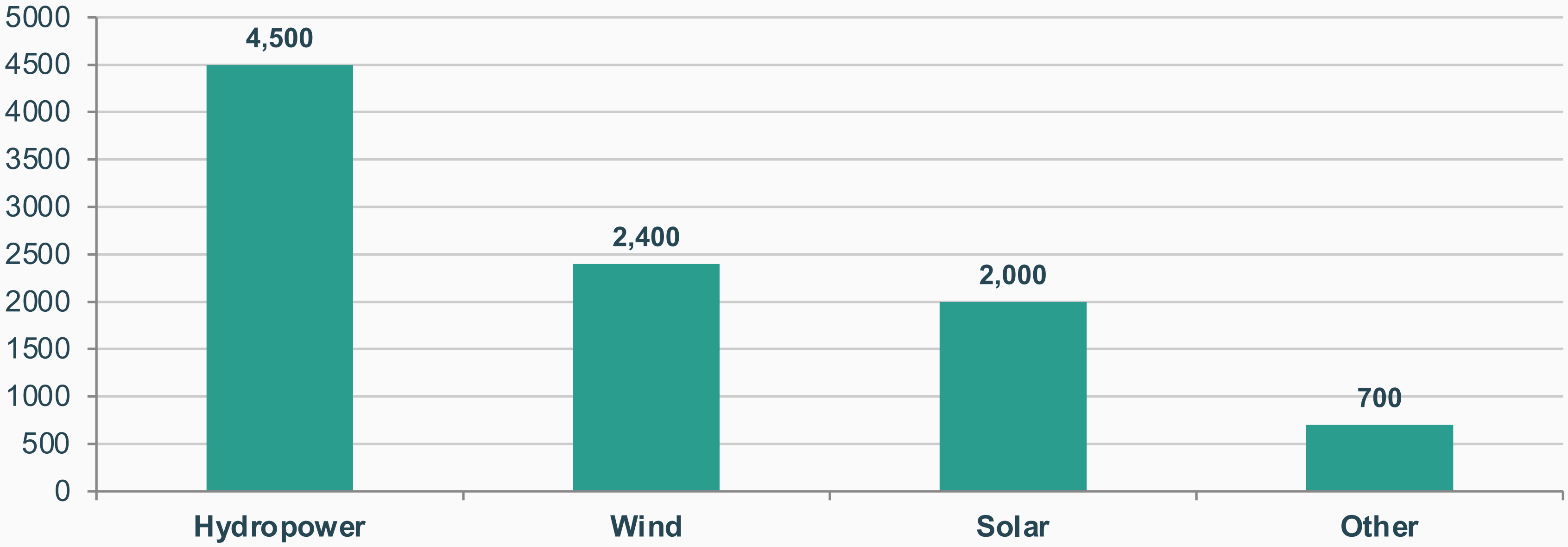
of global primary energy comes from renewables (2024, substitution method)

Primary energy = electricity + transport + heating combined



Source: Our World in Data, Ember (2024)

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



Wind + solar combined (~4,400 TWh) now nearly match hydropower (~4,500 TWh)

Solar is the fastest-growing source, with dramatic growth since 2010

Source: Our World in Data, Ember (2024)

Almost One-Third of Global Electricity Now Comes from Renewables

Regional leaders in renewable electricity share:

Country/Region	Renewable Share of Electricity
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%
World	~30%

Several African & Middle Eastern countries remain below 10%

Source: Our World in Data, Ember (2025)

Hydropower Remains the Largest Renewable Source at ~50% of Generation

50%

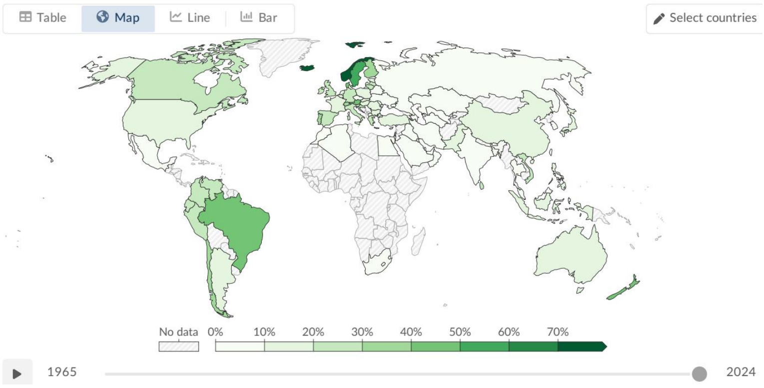
~4,500 TWh of renewable electricity from hydropower

Top Producers:

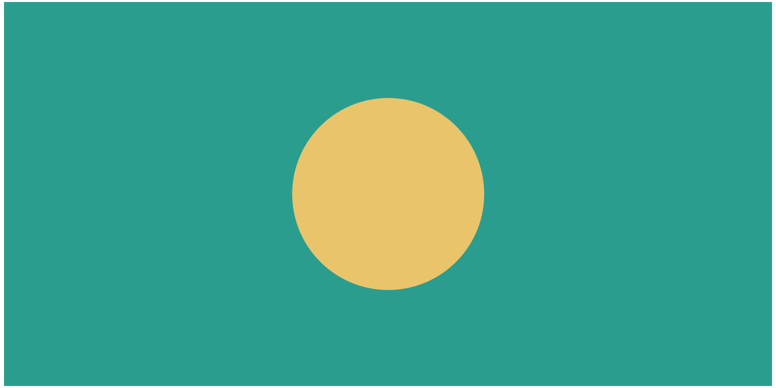
China, Brazil, Canada

Source: Our World in Data, Ember (2025)

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind: ~2,400 TWh (~25% of renewable electricity)



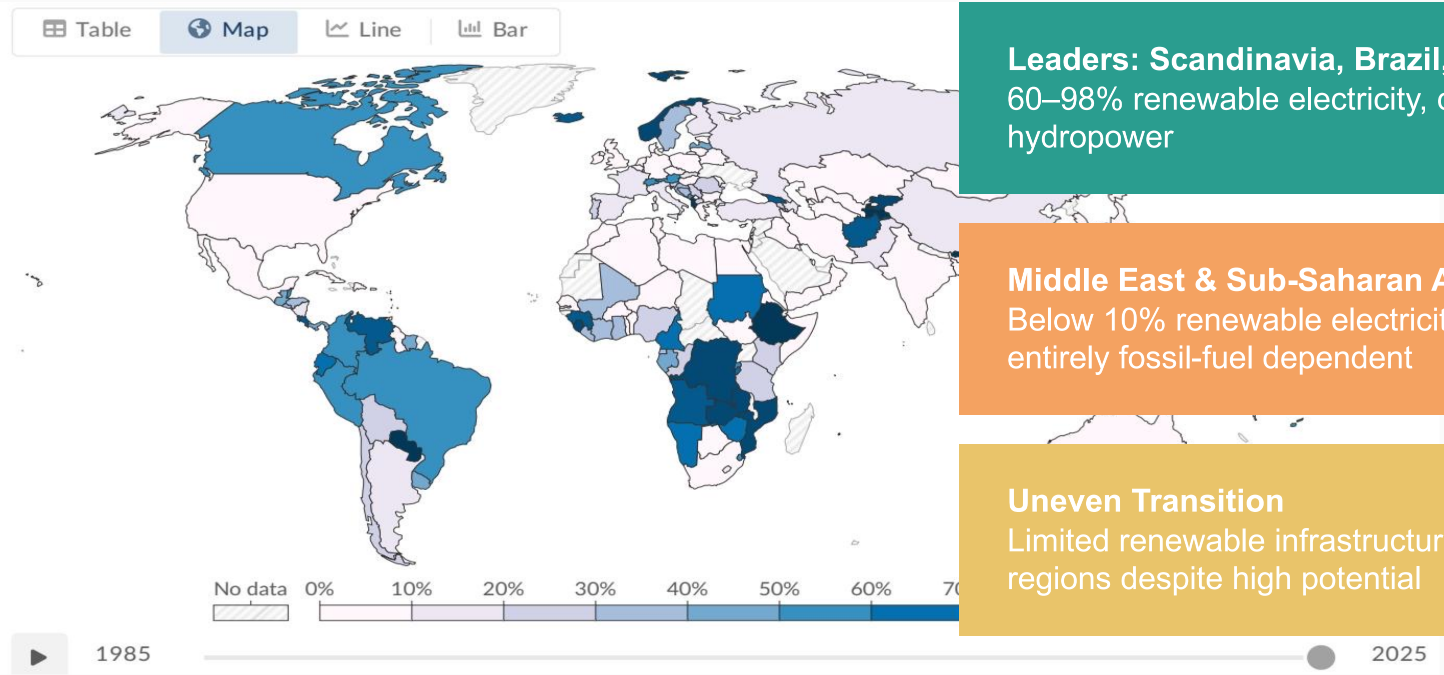
Solar: ~2,000 TWh (~21% of renewable electricity)

Fastest-growing source since 2010, driven by falling costs and policy support

Wind + Solar combined (~4,400 TWh) ≈ Hydropower (~4,500 TWh)

Source: Our World in Data, Ember (2024)

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Leaders: Scandinavia, Brazil, Canada
60–98% renewable electricity, driven by hydropower

Middle East & Sub-Saharan Africa
Below 10% renewable electricity, almost entirely fossil-fuel dependent

Uneven Transition
Limited renewable infrastructure in developing regions despite high potential

Source: Our World in Data, Ember (2025)

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

- Renewables at ~14% of primary energy and ~30% of electricity — a milestone but far from sufficient for climate targets
- Hydropower remains dominant (~50% of renewable electricity) but growth potential is limited vs. wind and solar
- Wind and solar are the fastest-growing technologies in history; combined, they now nearly match hydropower output
- Stark regional disparities: 60–90%+ renewable electricity in Scandinavia/Latin America vs. below 10% in Middle East/Sub-Saharan Africa
- Transport and heating lag far behind electricity in the renewable transition

The transition must accelerate to meet climate targets

Source: Our World in Data | Data: Ember, Energy Institute (2025)